

Dynamic Display Using LED Strip

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Surface-mount device (SMD) LEDs are becoming popular because of their low power consumption and high brightness. These LEDs are readily available in the market in many colours both as single LEDs and LED strips.

Here we describe a microcontroller-based dynamic display system using LED strip that can be used for advertising message display. The display works by flashing characters sequentially and then the entire text together. Different lighting effects are possible, like flashing in zig-zag, increasing, decreasing and random patterns.

LED strip

An LED strip is made up of a number of SMD LEDs affixed to a PCB

surface. LED strips are mainly of two types—flexible and rigid. These come in different lengths and light outputs.

The type of LED strip can be selected as per the desired application:

Bare strip. This LED strip is ideal for indoor (inside or under cupboards and stairs) and indirect lighting applications.

Silicon-covered strip. This LED strip is suitable for use in humid places like bathrooms and for outdoor use (accent lighting in houses and buildings).

Silicon-sleeved LED strip. This strip is fully encapsulated by a silicon sleeve and is suitable for use in swimming pools, saunas and outdoors where direct contact with rain or water is unavoidable.



The LED strip has in-built series dropper resistors and works off +12V, 400mA DC supply. It can be cut or trimmed to a smaller size of about 5 cm in length, and pasted on any flat surface (LED strips come with a self-sticking gum). In this dynamic display, we have cut the LED strip to form the word 'EFY'.

Circuit description

Fig. 1 shows the circuit for microcontroller-based dynamic display using LED strip while its power supply is shown in Fig. 2. It comprises micro-

controller AT89C51 (IC1), octal peripheral driver array ULN2803 (IC2, IC3, IC4 and IC5), regulator IC 7805 (IC6) and a few discrete components.

Microcontroller AT89C51 is at the heart of the circuit. It is an 8-bit microcontroller with 4 kB of Flash programmable and erasable read-only memory (PEROM), 128 bytes of RAM, 32 input/output (I/O) lines, two 16-bit timers/counters, a five-vector two-level

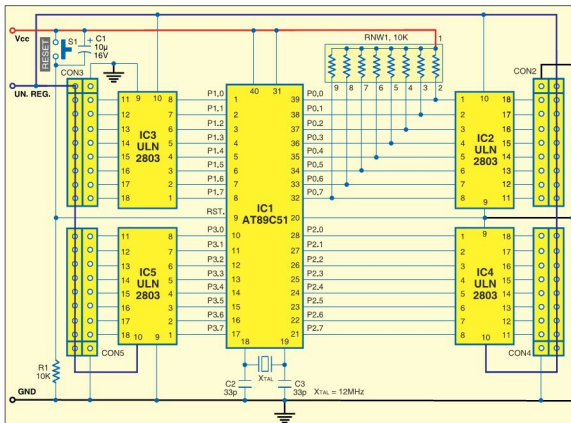


Fig. 1: Circuit for microcontroller-based dynamic display using LED strip